NATUREX

PANCHA BHOOTHALU • FIVE-ELEMENT ENVIRONMENTAL MONITORING

THEME: SUSTAINABLE DEVELOPMENT

An IoT-based environmental monitoring prototype inspired by Earth, Water, Fire, Air and Space. Designed to observe environmental conditions, identify abnormal readings and support timely action.

PROJECT DETAILS	TECHNOLOGY
Project: NATUREX – Pancha Bhoothalu **Team:** Team Snippers	**Controller:** ESP32 **Simulation:** Wokwi ESP32 Simulator
Cloud: ThingSpeak	**Programming:** C/C++ (Arduino framework)

FIVE ELEMENTS. ONE MISSION.

Monitoring the Five Elements, Protecting Our Environment.

README

Project overview

NATUREX is an IoT-based environmental monitoring prototype inspired by **Pancha Bhoothalu** (the five elements of nature): Earth, Water, Fire, Air and Space. An ESP32-based system collects environmental readings, evaluates them against predefined demonstration thresholds and provides status updates through a centralized dashboard.

NATUREX monitors soil moisture, water pH, temperature, humidity, air quality, light intensity, water-tank level and movement. It classifies conditions as **Normal**, **Watch** or **Alert**. Depending on programmed conditions, the prototype can activate an alarm and an automatic irrigation pump.

Prototype: Wokwi ESP32 Simulator | Cloud: ThingSpeak | Language: C/C++ (Arduino framework)

1. The Five Elements of Nature

NATUREX maps environmental measurements to the five elements as follows:

ELEMENT	MONITORING PARAMETERS	PURPOSE
Earth	Soil moisture and vibration	Identifies dry or waterlogged soil conditions and detects vibration events.
Water	Water pH and tank level	Identifies abnormal pH readings and low water levels.
Fire	Temperature and humidity	Monitors temperature changes and potential high-temperature/fire-risk conditions.
Air	Air quality (MQ135)	Monitors air-quality sensor readings and identifies potentially poor conditions.
Space	Light intensity (LDR) and motion (PIR)	Monitors surrounding visible-light conditions and movement.

Note: Space monitoring in the current prototype represents visible-light and motion monitoring. UV and radiation sensing are future enhancements; they are not measured by the present setup.

2. How NATUREX Works

The ESP32 acts as the central processing unit. Sensors provide readings to the controller, which processes values and compares them with programmed thresholds.

• Sensors collect environmental readings.

• The ESP32 processes the collected values.

• The system compares readings with predefined threshold values.

• Conditions are classified as Normal, Watch or Alert.

• LED indicators, an alarm relay and an irrigation pump respond according to programmed conditions.

• Readings can be uploaded to ThingSpeak for cloud-based monitoring.

System architecture

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Environmental Sensors → ESP32 Controller → Sensor Data Processing → Condition Classification → Normal / Watch / Alert
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The classified status can drive LED indicators, the alarm system and the automatic irrigation pump. Processed readings can also be sent to ThingSpeak for charts viewable on a laptop or mobile device.

Alert levels

LEVEL	MEANING
0 – Normal	Environmental conditions are within the defined normal range.
1 – Watch	Conditions require attention.
2 – Alert	A critical condition has been detected according to prototype thresholds.

The overall system status is determined by the highest alert level among the monitored parameters.

3. Prototype Simulation

The complete prototype is designed in the Wokwi ESP32 Simulator. This makes it possible to demonstrate system logic without requiring all physical sensors and hardware during the simulation stage.

COMPONENT	MONITORING / CONTROL FUNCTION	ESP32 PIN
Potentiometer 1	Soil-moisture simulation	GPIO 35
Potentiometer 2	Water-pH simulation	GPIO 34
Potentiometer 3	Air-quality simulation	GPIO 36
DHT22	Temperature and humidity	GPIO 25
Photoresistor (LDR)	Light intensity	GPIO 39
Ultrasonic sensor	Water-tank level	GPIO 14, 12
PIR sensor	Motion detection	GPIO 13
Push button	Vibration simulation	GPIO 18
Blue LED	Vibration indication	GPIO 16
Red LED	Watch and Alert indication	GPIO 17
Relay 1	Alarm system	GPIO 26
Relay 2	Irrigation pump	GPIO 27

Example simulation tests

• **Water:** Adjust the pH potentiometer to approximately 4.6 to simulate abnormal water pH.

• **Earth:** Reduce soil moisture to approximately 10% to activate the irrigation pump.

• **Fire:** Increase temperature to 65°C to demonstrate a high-temperature alert.

• **Air:** Increase the air-quality potentiometer value to simulate poor air quality.

• **Space:** Change light intensity and activate the PIR sensor to demonstrate environmental monitoring.

• **Vibration:** Press the push button to simulate vibration detection.

• **Water tank:** Adjust ultrasonic distance to simulate a low water level.

4. Cloud Monitoring – ThingSpeak

NATUREX is designed to support cloud-based monitoring through ThingSpeak. The ESP32 firmware is intended to send environmental readings to two ThingSpeak channels.

CHANNEL	FIELDS
Channel 1 – NATUREX Main	1. Temperature 2. Humidity 3. Air Quality 4. Light Level 5. Tank Distance 6. Water pH 7. Soil Moisture 8. Motion
Channel 2 – NATUREX Vibration	1. Device ID 2. Vibration Status

Cloud integration process

• Create an account on ThingSpeak.

• Create two channels using the fields listed above.

• Obtain the Channel IDs and Write API Keys.

• Configure the channel credentials in the ESP32 firmware.

• Run the simulation and verify that readings are uploaded.

• Open the ThingSpeak charts in a laptop or mobile browser.

Security: Cloud uploading requires valid ThingSpeak channel credentials and a working network connection. Do not publish private API keys in a public GitHub repository.

5. Environmental Alert Conditions

The following are demonstration thresholds used by this prototype. They are not official environmental, agricultural or safety standards. Values should be calibrated and validated before real-world deployment.

PARAMETER	NORMAL	WATCH	ALERT
Water pH	6.5–8.5	6.0–6.5 or 8.5–9.0	Below 6.0 or above 9.0
Soil moisture	35–65%	20–35% or 65–80%	Below 20% or above 80%
Temperature	10–38°C	Below 10°C or 38–50°C	50°C and above
Rapid temperature rise	No rapid rise	Rise of 5°C within 25 seconds	High-temperature alert at 50°C and above
Air-quality index*	Below 100	100–200	200 and above
Tank level	Distance below 30 cm	Distance of 30 cm or more, or no echo	Not separately defined
Vibration	No vibration	Vibration detected	Not separately defined

*The MQ135 is a gas sensor; its raw output is not automatically an official AQI measurement. AQI-like values should be treated as indicative simulation values unless a validated conversion and calibration method is implemented.

The automatic irrigation pump activates when soil moisture falls below the programmed critical threshold and switches off when moisture rises above the recovery threshold of approximately 45%.

6. Real-World Hardware Implementation

The simulated design can be adapted for physical implementation using an ESP32 development board and suitable sensors. Suggested hardware:

MONITORING / CONTROL FUNCTION	SUGGESTED HARDWARE
Water quality	pH sensor with PH-4502C interface
Soil moisture	Capacitive soil-moisture sensor
Air quality	MQ135 gas sensor
Temperature and humidity	DHT22
Light intensity	LDR module
Water-tank level	HC-SR04 ultrasonic sensor
Motion	PIR sensor
Vibration	SW-420 vibration sensor
Control	ESP32 development board
Output	LEDs, relay module, alarm and pump

Hardware safety considerations

• ESP32 GPIO pins use 3.3V logic. Protect GPIO inputs from signals above the permitted voltage.

• Use suitable voltage dividers or level-shifting circuits for sensor outputs that may exceed 3.3V.

• Calibrate the pH sensor using appropriate buffer solutions.

• MQ135 readings require warm-up and calibration; they should not be treated as precise measurements without validation.

• Use suitable current-limiting resistors with LEDs.

• Use appropriate relay drivers and electrical isolation for external loads. Mains-powered equipment should be handled by a qualified person.

7. Prototype Limitations

• The current project uses Wokwi simulation rather than physical sensor measurements.

• Potentiometers simulate water pH, soil moisture and air-quality readings.

• A push button simulates vibration detection.

• LDR readings represent visible-light intensity, not UV or radiation measurements.

• MQ135 readings are approximate and require calibration for practical applications.

• Alert thresholds are demonstration values and require validation before real-world deployment.

• Actual cloud monitoring depends on valid ThingSpeak credentials and network connectivity.

8. Future Enhancements

• Integrate UV and radiation sensors for expanded Space monitoring.

• Add a water-turbidity sensor for improved water-quality assessment.

• Introduce mobile notifications for critical environmental conditions.

• Explore solar-powered operation for outdoor monitoring.

• Improve sensor calibration and real-time data analysis.

• Provide agricultural recommendations based on soil and water conditions.

• Expand the cloud dashboard with historical readings and environmental trends.

9. Applications

• Smart agriculture and irrigation management.

• Environmental monitoring in campuses and community spaces.

• Water-quality observation and tank-level monitoring.

• Air-quality monitoring demonstrations.

• Industrial environmental monitoring and safety demonstrations.

• Sustainable-development and IoT learning projects.

10. Project Information

ITEM	DETAILS
Project name	NATUREX – Pancha Bhoothalu
Theme	Sustainable Development
Project category	IoT and Environmental Monitoring
Controller	ESP32
Simulation platform	Wokwi ESP32 Simulator
Cloud platform	ThingSpeak
Programming language	C/C++ (Arduino framework)
Developed by	Team Snippers

Acknowledgements

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NATUREX – Monitoring the Five Elements, Protecting Our Environment.

Suggested GitHub repository tagline: *Five Elements. One Mission. A smarter approach to environmental monitoring.*